Location and Functions of Key Structures

1. Excretory System

S. No.	Structure	Location	Function
1	**Kidney/Nephron**	*Lie* along the *posterior abdominal wall*, *one on either side* of the *vertebral column*.	*Excretion* of *Nitrogenous wastes* like *urea*. *Osmoregulation*.
2	**Ureters**	They arise from the *renal pelvis* of the *kidneys* at the *hilum* and the other end opens into the *urinary bladder*.	*Transports urine* from the *pelvis* of the *kidney* to the *urinary bladder*.
3	**Urinary bladder**	*Sac like structure* situated in the *lower end* of the *abdomen* into which the other end of the ureter opens.	*Temporary storage* of *urine*.
4	**Urethra**	*Short, muscular tube* arising from the *neck* of the *urinary bladder* and leading to the *outside*.	Allows the *passage of urine* from *urinary bladder* to *outside* the body.
5	**Glomerulus**	*Knot of capillaries* present in the *cup* of the *Bowman's capsule*.	Brings about *ultrafiltration*.
6	**Proximal convoluted tubule**	*Lies* in the *cortex* of the *kidney*.	*Reabsorbs water, glucose* and *sodium and chloride ions* from *glomerular filtrate* into *renal capillaries*.
7	**Loop of Henle**	*Present* in the *renal medulla*.	*Reabsorption* of *water* and *sodium ions* from *glomerular filtrate* to *renal capillaries*.
8	**Distal convoluted tubule**	*Present* in the *cortex* of the *kidney*.	*Tubular secretion* of *K+, Creatinine* and *foreign substances* like *antibiotics* into urine.
9	**Collecting duct**	*Formed by the union* of *collecting tubules*, passes through the *cortex, medulla* and opens into the *pelvis* of the *kidney*.	*Transports urine* from *nephron* to the *pelvis* of the *kidney*.

2. Circulatory System

S. No.	Structure	Location	Function
1	**Heart**	*Situated between* the *two lungs* in the *thoracic cavity*. The *apex* is *tilted towards the left side*.	*Pumps oxygenated blood* from *lungs* to the *different parts* of the *body*. *Pumps deoxygenated blood* from *parts of body* to *lungs* for *oxygenation*.
2	**Pericardial fluid**	Found in the *pericardial cavity*, *between* the *two membranes* of the *pericardium*.	*Protects* the *heart* from *mechanical injury* and *shock*. Acts as a *lubricant* and *reduces friction* for the *beating* of the *heart*.
3	**Sino atrial node (SA node) or Pacemaker**	*Upper wall* of *right atrium* close to the *opening* of the *vena cava*.	*Initiates* and *regulates* the *heart beat*. *Receives impulse* from *medulla oblongata* and *regulates* the *contraction* of the *auricles*.
4	**Atrio Ventricular node (AV node)**	*Bottom* of the *right auricle* near the *tricuspid valve*.	*Regulates* the *contraction* of the *ventricles*.
5	**Bundle of His**	*Arises* from the *AV node* and *extends* to the *interventricular septum*.	It *receives an impulse* from the *AV node* and *transmits* it to the *Purkinje fibres*.
6	**Bicuspid valve or mitral valve**	*Between* the *left auricle* and *ventricle* of the *heart*.	*Prevents* the *backflow of blood* from the *left ventricle* to the *left auricle*.
7	**Tricuspid valve**	*Between* the *right auricle* and *ventricle* of the *heart*.	*Prevents* the *backflow of blood* from the *right ventricle* to the *right auricle*.
8	**Chordae tendinae**	*Extend* from the *papillary muscles* to the *pointed ends* of the *cuspid valves*.	*Keeps* the *tricuspid* and *bicuspid valves* in *position* and *prevents* them from *upturning* into the *auricles*.
9	**Pulmonary semilunar valve**	At the *opening* of the *pulmonary artery* which arises from the *right ventricle*.	*Prevents backflow* of *deoxygenated blood* from the *pulmonary artery* into the *right ventricle*.
10	**Aortic semilunar valve**	At the *opening* of the *aorta* which arises from the *left ventricle*.	*Prevents backflow* of *oxygenated blood* from the *aorta* into the *left*

			ventricle.
11	**Coronary artery**	*Arises* from the *aortic arch*, near its *base* as it emerges from the *heart*.	*Supply oxygenated blood* to the *heart muscle*.
12	**Spleen**	*Behind* the *stomach* above *left kidney*.	*Produces lymphocytes, destroys RBC* etc.
13	**Valves in veins**	*Veins*.	*Prevent backflow of blood* since *blood flows* to the *heart against gravity*.

3. Endocrine System

S. No.	Structure	Location	Function
1	**Adrenal gland**	*Attached like a cap above each kidney*.	
1.a	**Adrenal Cortex**	*Outer region* of the *adrenal gland*.	*Secretes corticoids - Glucocorticoids, Mineralocorticoids, Sex corticoids*.
1.b	**Adrenal medulla**	*Inner region* of the *adrenal gland*.	*Secretes adrenaline* which *increases heartbeat, increases blood pressure, raises blood sugar level*, preparing body to *cope* with *emergency situations*.
2	**Thyroid gland**	*Front* of the *neck*, *just below* the *larynx*.	*Secretes thyroxine* which *regulates* the *basal metabolic rate* of the *body*.
3	**Pancreatic gland**	*Behind* the *stomach*, at the *loop* of the *duodenum*.	*Secretes hormones* like *Glucagon* and *Insulin*. *Glucagon increases blood sugar level* by promoting *conversion* of *glycogen* to *glucose*. *Insulin decreases* the *blood sugar level* by promoting *conversion* of *glucose* into *glycogen*.
4	**Pituitary gland**	*Hanging* from *base* of the *midbrain below* the *hypothalamus*.	*Anterior lobe* of *pituitary secretes hormones* like *growth hormone* and *tropic hormones* such as *TSH, ACTH, LH, FSH* (only full forms to be written). *Posterior lobe* of *pituitary secretes hormones* like *Oxytocin* and *Vasopressin (ADH)*.

4. Nervous System

4.1 Parts of Neuron

S. No.	Structure	Location	Function
1	**Perikaryon/Cyton/Cell body**	*Proximal part* of a *neuron*; Structure: *centrosome absent*.	*Receives nerve impulses* from *axonal terminals* of *another neuron* and *conducts* it to the *cyton* of its *own neuron*.
2	**Dendrites**	*Cytoplasmic projections* of *cyton/perikaryon*.	*Transmits stimulus* to *cyton* of *neuron*.
3	**Axon**	*Long process* of the *cell body*.	*Conducts nerve impulses away* from the *cyton* to the *axon endings*.
4	**Myelin sheath**	*Covering/surrounding* the *axon*.	*Prevents* the *mixing of impulses* between *2 adjacent neurons*. *Increases* the *transmission* of *nerve impulse conduction* along the *axon*.
5	**Nodes of Ranvier**	*Unmyelinated gaps between* the *myelin sheath*.	*Increases* the *speed* of *nerve impulse conduction*.
6	**Neurilemma**	*Surrounding* the *myelin sheath*.	*Protects* the *neuron*.
7	**Terminal branches/axon terminals**	*Posterior end* of the *axon*.	*Receives nerve impulses* conducted through the *axon* and *transmits* it to the *next neuron*.
8	**Synaptic bulb**	*Distal end* of the *terminal branches*.	*Contains chemicals* called *neurotransmitters* (e.g., *acetylcholine*). When the *nerve impulse reaches* the *terminal branches, neurotransmitters are released* by the *synaptic bulb*.
9	**Neurotransmitters**	*Found* in the *synaptic bulb*.	*Released* into the *synapse* by the *synaptic bulb*. Hence these *chemicals* are *responsible* for *transmission* of *nerve impulses* from the *terminal branches* of one neuron to the *dendrites* of the *next neuron*.
10	**Synapse**	Is the *junction* between the *terminal branches* of one neuron and the *dendrites* of the *adjacent neuron*.	*Neurotransmitters* are *released* into the *synapse*, which *sets off a wave of excitement* in the *next neuron*.

4.2 Types of Neuron

S. No.	Structure	Location	Function
1	**Sensory neuron**	*Extends* from *receptors (sense organs)* to *Central Nervous System*.	*Transmits nerve impulses* of *stimuli received* from the *sense organs (receptors)* to *brain* or *spinal cord*.
2	**Motor neuron**	*Extends* from the *Central Nervous System* to the *effectors (muscles and glands)*.	*Transmits nerve impulses* for *response* from *CNS* to the *muscles and glands (effectors)*.
3	**Association neuron**	*Found* in the *spinal cord* and *brain*.	*Interconnect* the *sensory neuron* and the *motor neuron*.

4.3 Part of Brain

S. No.	Structure	Location	Function
1	**Cerebrum**	*Part* of the *forebrain* and is the *largest part* of the *brain*.	It is the *seat of intelligence, memory, thinking, reasoning* and *emotions*. *Centre* for *perception* of *stimuli* for *vision, smell, taste* and *hearing*. All *voluntary activities* are *initiated* in the *cerebrum*.
2	**Corpus callosum**	*Transverse band* of *nerve fibres*, *internally joining* the *two cerebral hemispheres*.	*Transmits nerve impulses* from the *right cerebral hemisphere* to the *left hemisphere* and *vice versa*.
3	**Diencephalon** (Hypothalamus and Thalamus)	*Between* the *two cerebral hemispheres*, just *superior* to the *midbrain*.	**Hypothalamus:** *Controls pituitary gland, sleeping and waking patterns, hunger* and *thirst*. Also *regulates body temperature*. **Thalamus:** *Interprets sensory impulses* such as *pain, touch* and *pressure*.
4	**Mid brain**	*Between* the *anterior* and *posterior part* of the *brain*.	It *relays sensory impulses* from the *spinal cord* to the *thalamus*. It *relays motor impulses* from the *thalamus* to the *spinal cord*.
5	**Hind brain**		
5.a	**Cerebellum**	*Located under* the *cerebrum*.	*Maintains* the *body balance* and *coordinating* the *movements* of the *skeletal muscles*.
5.b	**Pons**	*Between* the *two lobes* of	*Transmits nerve impulses* across the *two lobes* of

		the *cerebellum* on the *ventral side*.	the *cerebellum*, thus *ensuring a coordination* of *muscular movements* on *both sides* of the *body*.
5.c	**Medulla oblongata**	*Below* the *cerebellum*.	*Controls* the *involuntary activities* of *internal organs* such as *breathing, heart beat* and *peristaltic movements* of the *alimentary canal*.

5. Male Reproductive System

S. No.	Structure	Location	Function
1	**Testes**	*Contained* in the *thin walled scrotum*.	*Produces sperms* & *Hormone Testosterone*.
2	**Seminiferous tubules**	*Present* in the *lobules* of the *testes*.	*Spermatogenesis* (*Produces sperms*).
3	**Interstitial Cells/Leydig Cells**	*Present* as *packing tissue* in the *lobules* of the *testes*, *between* the *seminiferous tubules*.	*Produces* the *male hormone Testosterone*.
4	**Epididymis**	*Like a cap* on the *upper pole* of *each testis*.	*Stores* the *sperms* till they *mature* & *become motile*.
5	**Vas Deferens**	*Connects* the *epididymis* to the *urethra*.	*Transports sperms* from the *epididymis* to the *urethra*.
6	**Penis**	*Lies in front* of the *scrotum*.	*Serves* for *passing out* of *both semen* and *urine*.
7	**Seminal Vesicles**	*Between posterior surface* of *urinary bladder* & *rectum*.	*Produces* an *alkaline secretion* which *activates/provides motility* to the *sperms*.
8	**Prostate Gland**	*Surrounds* the *Urethra*, at the *base* of the *Urinary Bladder*.	To *produce* an *alkaline secretion* to *neutralise* the *acidic vagina*.
9	**Bulbourethral Gland or Cowper's gland**	*Located beneath* the *prostate gland* on *either side* of the *urethra*.	Its *alkaline secretion lubricates* the *end* of the *penis* during *coitus*.
10	**Acrosome**	*Top* of the *head* of the *sperm*.	*Releases hyaluronidase* to *dissolve* the *walls* of the *ovum* allowing *sperm nucleus* to *enter* the *ovum*.

6. Female Reproductive System

S. No.	Structure	Location	Function
1	**Graafian follicle**	*Ovary*.	
2	**Corpus luteum**	*Yellow remnants* of the *Graafian follicle* after *ovulation*.	*Secretes progesterone* to *maintain thickness* of *endometrium*.
3	**Ovaries**	*Upper pelvic cavity*, *one on each side* of *uterus*.	*Oogenesis, Ovulation* & *Secretion* of *Hormone Progesterone* & *Oestrogen*.
4	**Oviduct/Fallopian Tube/Uterine Tube**	*Extends laterally* on *each side* of the *uterus*.	*Site of fertilization*.
5	**Oviducal Funnel**	*Mouth* of *oviduct*.	*Receives* the *released ovum* and *pushes* it into the *oviduct*.
6	**Uterus**	*Pear-shaped, hollow, muscular organ* situated *between urinary bladder* and *rectum*.	*Protects* & *nourishes* the *developing embryo*.
7	**Endometrium**	*Innermost vascular layer* of the *uterus*.	*Thickens* to *receive* the *fertilized egg* and *gets discharged* during the *Menstrual Phase*.
8	**Cervix**	A *small lower constricted part* of the *uterus*.	*Enlarges* during *parturition*.
9	**Vagina**	A *muscular tube starting* from *lower end* of the *uterus* to the *outside*.	*Receives* the *penis* during *coitus* (*intercourse*).
10	**Amnion**	*Surrounds* the *embryo* except at the *thick umbilical cord*.	*Membrane surrounding* the *foetus* and the *amniotic fluid*.
11	**Amniotic Fluid**	A *fluid within* the *amnion*, *surrounding* the *foetus*.	*Protects* the *embryo* against *mechanical shocks* and *injury*. *Keeps even pressure* around the *embryo*. *Prevents sticking* of the *foetus* to the *amnion*.
12	**Chorion**	*Forms* the *Villi* of the *placenta* from the *foetal side*.	*Helps* in *diffusion* of *substances* from the *mother* to the *foetus* & *foetus* to *mother*.
13	**Allantois**	*Second layer* of the *foetal membranes*.	*Forms* the *umbilical cord*.

14	**Placenta**	A *disc like structure attached* to the *uterine wall*.	**Transport:** *transports oxygen* and *nutrients* from *maternal blood* to *foetus*; *transports carbondioxide* and *wastes* from the *foetus* to the *maternal blood*. **Protective:** *prevents pathogens* from the *maternal body* from *entering* into *foetus*; *prevents blood pressure* of *maternal blood* from *affecting* the *foetus*. Also *acts* as an *endocrine gland*.
15	**Umbilical Cord**	A *cord containing foetal blood vessels*.	*Connects* the *placenta* to the *foetus*.

7. Eye

S. No.	Structure	Location	Function
1	**Tear glands**	*Under* the *upper eyelids*, at the *upper, outer corner* of the *orbit* of the *eye*.	*Secretes tears - cleanses* the *eye* by *washing away foreign substances* that may *enter* the *eye*.
2	**Conjunctiva**	*Covering* of the *entire front surface* of the *eye*, *continuous* with the *inner eyelids*.	*Contains lysozymes* - this *protects* the *eye* from *infections*. *Protects* the *surface* of the *eye*.
3	**Sclera**	*Outermost white layer* of the *eyeball*.	*Maintains* the *shape* of *eyeball*.
4	**Cornea**	*Anterior, bulging* and *transparent part* of *sclera*.	*Refracts* the *light rays* as they *pass through it*.
5	**Choroid**	*Middle layer* of the *eye*, *inner* to the *sclera*.	As it is *pigmented*, it *absorbs light rays* and *prevents* the *reflection* of the *light rays*. *Blood vessels* present here, *nourish* the *eye*.
6	**Iris**	*Located between* the *lens* and the *cornea* on *extension* of the *choroid*.	*Changes* the *size/diameter* of the *pupil*.
7	**Pupil**	*Opening seen* in the *center* of the *iris*.	*Regulates* the *amount of light entering* into the *eye*.
8	**Ciliary body/muscle**	At the *junction* of the *choroid* and *iris*.	*Ciliary processes secrete aqueous humour*. *Ciliary muscles alter* the *shape* of the *eye lens* for *accommodation*.

9	**Suspensory ligaments**	*Extends* from the *ciliary muscles* to the *eye lens*.	*Holds* the *eye lens* in *position*.
10	**Retina**	*Innermost layer* of the *eyeball*.	It is the *photosensitive layer* of the *eye*. *Contains rods* and *cones*.
11	**Rods** (contains the pigment Rhodopsin)	*Present* at the *periphery* of the *retina*.	*Respond* to *low light intensity* and so *enable vision* in *dim light*.
12	**Cones** (contain a pigment called Iodopsin)	*Present mainly* at the *yellow spot*.	*Responsible* for *colour vision* and for *vision* in *bright light*.
13	**Optic nerve**	*Leaves* the *eye* at the *blind spot* and *reaches* the *occipital centre* of the *brain*.	*Transmits nerve impulses* from the *sensory cells* (*rods* and *cones*) of the *retina* to the *visual area* (*occipital region*) of the *cerebrum*.
14	**Lens**	*Located behind* the *iris* and *pupil*, *supported* by the *suspensory ligaments*.	*Focuses* the *light rays* on the *retina*.
15	**Aqueous humour**	In the *anterior chamber*, *between* the *lens* and the *cornea*.	*Keeps* the *lens* and *cornea moist*. *Protects* the *lens* from *mechanical shock* and *injury*.
16	**Vitreous humour**	In the *vitreous chamber*, *between* the *lens* and the *retina*.	*Maintains* the *shape* of the *eyeball*. It *refracts* the *light rays* before it *reaches* the *retina*.
17	**Yellow spot**	At the *back* of the *eye*, *in line* with the *horizontal axis* of the *eyeball*.	It is the *region of best vision* as it *contains* the *highest concentration* of *cones*.
18	**Blind spot**	*Opening lateral* to the *yellow spot*, where the *optic nerve leaves* the *eyeball*.	This is an *area of no vision*, since *rods* and *cones* are *absent* here.

8. Ear

S. No.	Structure	Location	Function
1	**Ear pinna**		*Collects sound waves* and *directs* it into the *auditory canal*.
2	**Auditory canal**	*Outer ear*.	*Transmits sound waves* from *pinna* to the *ear*

			drum/tympanum.
3	**Tympanum/ear drum**	*Outer ear*.	*Converts sound waves* to *sound vibrations*.
4	**Ear Ossicles**	*Present* in *middle ear*.	They *amplify* the *sound vibrations received* from the *external ear* and *transmit* these to the *cochlea*.
4.a	**Malleus**	*Handle rests* on the *tympanum* and the *head rests* on the *incus*.	(*Hammer shaped*)
4.b	**Incus**		(*Anvil shaped*) is the *central small bone*.
4.c	**Stapes**	Its *base* is *attached* to the *oval window*.	(*Stirrup shaped*)
5	**Eustachian tube**	*Connects* the *pharynx* to the *cavity* of the *middle ear*.	It *equalizes* the *air pressure* on *either sides* of the *ear drum* so that it can *vibrate freely*.
6	**Oval window**	On the *inner boundary* of the *middle ear* and *inner ear*.	The *sound vibrations* from the *stapes* (*middle ear*) is *transmitted through* the *oval window* to the *cochlea* of the *inner ear*.
7	**Round window**	*Below* the *oval window* in the *inner ear*.	*Dissipates excess sound vibrations*.
8	**Cochlea**	*Inner ear* (*Membranous labyrinth*). Structure: *Bony canal* made of *two and a half spirals* in the *inner ear*. Function: *lower portion* of the *labyrinth*.	*Contains* the *Organ of Corti* which is *responsible* for *hearing*.
9	**Organ of Corti**	In the *basilar membrane* of the *middle canal/median canal/cochlear canal* of the *cochlea*.	*Transforms sound vibrations* into *nerve impulses*.
10	**Semicircular canals**	*Set of three fluid filled canals arranged* at *right angles* to *each other*. *Found* in the *upper portion* of the *labyrinth*.	*Contains sensory cells responsible* for *dynamic balance*.
11	**Utriculus and Sacculus (vestibule)**	*Between semicircular canals* and *cochlea* in *inner ear*.	*Contains sensory cells responsible* for *static balance*.
12	**Auditory nerve**	*Extends* from the *inner ear* to the *temporal lobe* of the *cerebrum*.	*Transmits auditory impulses* from the *cochlea* to the *cerebrum*.

9. Plant Structures

S. No.	Structure	Location	Function
1	**Lenticels**	On the *surface/bark* of *old, woody stems*.	*Exchange/diffusion* of *respiratory gases*.
2	**Cuticle**	*Wax like covering* on *epidermis* of *leaves* & *green stems*.	*Prevents excess loss of water* due to *Transpiration*.
3	**Mesophyll Cells**	*Below* the *epidermal cells* of the *dorsiventral leaf*.	*Site* of *photosynthesis*.
4	**Cambium**	*Below* the *bark* of the *tree*.	*Increases* the *girth* of the *stem*.
5	**Xylem**	*Veins* of *leaf, stem* and *leaves/ Innermost part* of *Vascular bundles*.	*Conducts water* and *minerals* (*sap*) from the *roots* to the *aerial parts* of the *plant*.
6	**Phloem**	*Found* in the *veins* of *leaf, stem* and *leaves/ Periphery* of the *Vascular Bundles*.	*Translocates* the *prepared food* in the *leaves* to *all other parts/* the *storage organs* of the *plant*.
7	**Hydathodes**	*Special openings* at the *margin* of *veins* on the *leaf margin*.	*Guttation*.
8	**Guard Cells**	On *either side* of *stoma/surrounds* the *stoma*.	*Regulate* the *opening* and *closing* of the *stoma*.
9	**Chloroplast**	In the *cytoplasm* of *green plant cells*.	*Site* of *Photosynthesis*.
10	**Chlorophyll**	In the *walls* of *thylakoid/grana* of the *chloroplast*.	*Traps sunlight/light* for *photosynthesis*.
11	**Thylakoids/Grana**	In the *chloroplast*.	*Site* of *Light Reaction*.
12	**Stroma**	*Colourless ground substance present* in the *chloroplast*.	*Site* of *light independent reaction*.
13	**Stoma**	*Minute opening surrounded* by *guard cells occurring* in *large number* on the *lower epidermis/ventral surface* of *leaf*.	*Diffusion* of *photosynthetic* and *respiratory gases, Transpiration*.